Student names: . . . (please update)

Instructions: *This week, you will be installing and testing the simulation environment that you will be using for the final project. Make sure to install it as soon as possible and get help from the TAs if you encounter any issues before you officially start the project next week.*

Simulation environment installation

To install all the project requirements follow the instructions in the *README.md* file in the project_installation folder.

This will install the *MuJoCo* simulator and the *dm_control* package which are software maintained by Deepmind for running robot simulations. It will also install the *farms* packages developed at the Biorobotics Laboratory (BioRob) to simulate various robotic environments, see https://github.com/farmsim. If you are interested in knowing more about *MuJoCo*, you can find out more on https://mujoco.org.

IMPORTANT: Make sure you have activated and are using your virtual environment and its python interpreter that that you have created for this course.

NOTE: If you are unclear about the basic steps then refer back to Lab 0 documentation or ask for help to the TAs.

Examples

After the installation is complete, you can run an example based on next week's project to get you accustomed to the MuJoCo graphical interface. Run the example.py script that is available under `project_installation/`, from that folder. You will be able to see the Polymander robot (a salamander-like robot) crawling on the ground.

You can run it by calling:

```
>> python example.py
```

Important note to Mac users:

- You will need to use `mjpython` instead of `python`. More information can be found at `https://mujoco.readthedocs.io/en/stable/python.html#passive-viewer`.
- If some error regarding video recording appears, a fix could be to install ffmpeg via

  ```
  >> brew install ffmpeg
  ```

  If you still have issues contact one of the TAs for help.
- Possible fixes to the error: "RuntimeError: Cannot create a GUI FigureManager outside the main thread using the MacOS backend. Use a non-interactive backend like 'agg' to make plots on worker threads.". Add at the top of the example.py scipt the following lines:

  ```
  >> import matplotlib
  >> matplotlib.use("Agg")
  ```

Graphical User Interface Interaction

When you run the example, a Graphical User Interface (GUI) should launch, like the one shown in Figure 1

You can use the left mouse click to move around the scene and right mouse click to rotate the camera. You can also select a part of the robot by double left clicking on a part. Once selected, you can then interact with it by holding the CONTROL key and dragging with left or right click. Try it out for yourself to familiarise with the interface.

There are many keyboard shortcuts also available

- Press SPACE to toggle play/pause
- Press "!" / "^" to change speed factor (between zero (0) and backspace)
- Press w to toggle wireframe
- Press t to toggle transparency
- Press s to toggle shadows
- Press c to show collisions
- Press f to show collisions forces, combine with p to show friction/reaction
- Press b to show external forces (try in water later on)
- And many more...

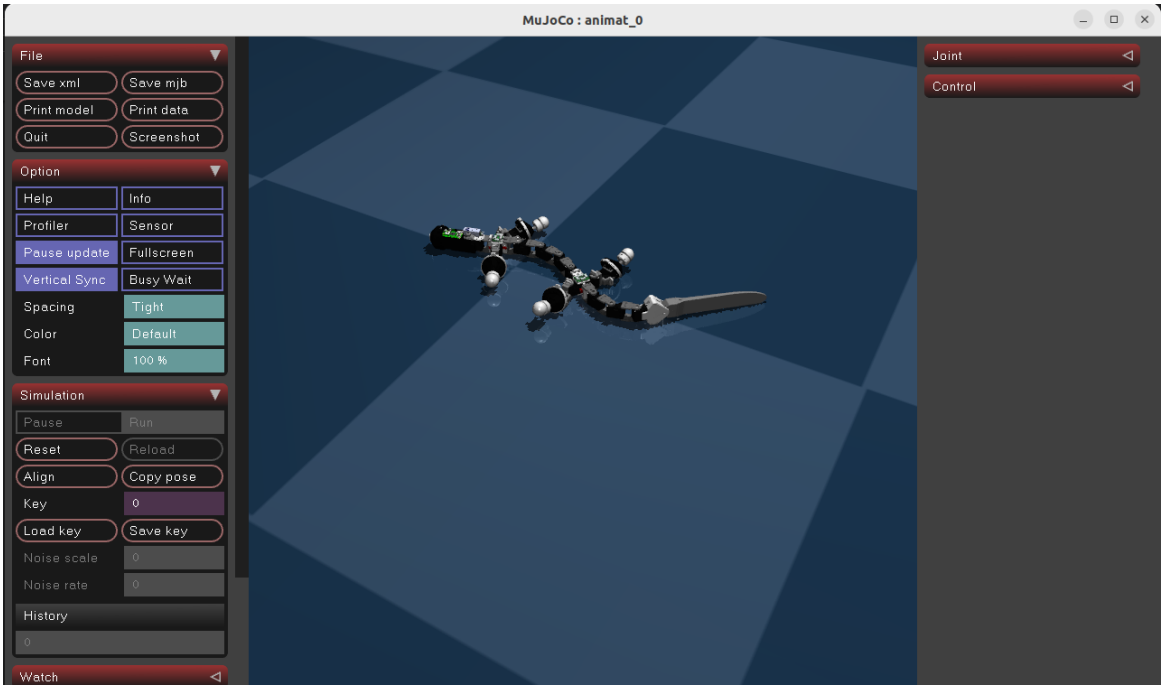

Figure 1: MuJoCo/dm_control Graphical User Interface

Important things to explore with the provided example

- Changing the view of the scene using the controls above
- Interaction with the objects in the scene
- Try changing to a water arena in the example script and showing the external forces acting on the body (press b to visualize the sum of buoyancy and drag forces)

Preparing for the project

Once you are done with the installation and have tried the simulation environment. The best way to prepare for the upcoming project is to read the following papers:

- Ijspeert, Auke Jan, et al. "From swimming to walking with a salamander robot driven by a spinal cord model." science 315.5817 (2007): 1416-1420.

- Thandiackal, Robin, et al. "Emergence of robust self-organized undulatory swimming based on local hydrodynamic force sensing." Science Robotics 6.57 (2021): eabf6354.
- Karakasiliotis, Konstantinos, et al. "From cineradiography to biorobots: an approach for designing robots to emulate and study animal locomotion." Journal of The Royal Society Interface 13.119 (2016).

Happy reading!